

# A Refutation of the Extremal Example in Conjecture 00000000036

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## Abstract

Conjecture 00000000036 asserts that a set  $A \subseteq [N]$  in which the sum (when  $\geq 4$ ) and the difference (when  $\geq 2$ ) of any two of its elements are both composite satisfies  $|A| \leq (1/3 + \varepsilon)N$ , with  $1/3$  attained by the residue classes  $\{\pm 1 \pmod 6\}$ . We show the extremal example is wrong: 5 and 7 both lie in  $\{\pm 1 \pmod 6\}$ , yet  $7 - 5 = 2$  is prime. Hence the conjecture is false as stated. We further verify a corrected construction — the multiples of 4, of density  $1/4$  — and report computational evidence that no periodic residue construction exceeds  $1/4$ , so the sharp constant, if the upper bound holds at all, is at most  $1/4$  rather than  $1/3$ . Both halves are formalized in Lean 4 against Mathlib.

## 1 Statement and reading

Conjecture 00000000036 concerns a set  $A \subseteq \{1, \dots, N\}$  such that

$$a, b \in A, \quad a \neq b \implies \begin{cases} a + b \geq 4 \implies a + b \text{ composite,} \\ |a - b| \geq 2 \implies |a - b| \text{ composite,} \end{cases} \quad (1)$$

and asserts

$$|A| \leq \left(\frac{1}{3} + \varepsilon\right) N \quad (2)$$

for every fixed  $\varepsilon > 0$  and all sufficiently large  $N$ , with equality asymptotically attained by

$$B = \{n \geq 1 : n \equiv \pm 1 \pmod 6\}. \quad (3)$$

Unlike some sibling conjectures, no distinctness repair is needed here: with  $a = b$  the sum  $2a$  is composite for  $a \geq 2$  and the difference 0 is exempt, so the hypothesis is non-degenerate as written. The quantifier over pairs is read, as usual, over distinct elements.

## 2 The claimed example fails

**Theorem 1.** *The set  $B$  in (??) does not satisfy (??). In particular the constant  $1/3$  in (??) is not attained by that construction, and Conjecture 00000000036 is false as stated.*

*Proof.*  $5 \pmod 6 = 5$  and  $7 \pmod 6 = 1$ , so  $5, 7 \in B$ . They are distinct, and  $|7 - 5| = 2 \geq 2$  with 2 prime. This single pair already violates (??).  $\square$

The failure is not a boundary effect: the offending pattern repeats in every block, since  $6k + 5$  and  $6k + 7$  both lie in  $B$  and differ by 2 for every  $k \geq 0$ . The residue-level explanation is instructive. Modulo 6 the difference is 2, with  $\gcd(2, 6) = 2 > 1$ , so a naive gcd test passes; what the test misses is that the realized difference *equals* the common divisor, which is itself prime. Any sound residue sieve must check small realizations, not just gcds.

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### 3 A corrected construction of density $1/4$

**Definition 2.** Let  $A = \{n \geq 1 : 4 \mid n\}$ , the positive multiples of 4.

**Proposition 3.**  $A$  satisfies (??).

*Proof.* Let  $a \neq b$  be distinct elements of  $A$ , both divisible by 4. Their sum  $a + b$  is divisible by 4, hence even; the smallest distinct positive multiples of 4 are 4 and 8, so  $a + b \geq 12 > 2$ , and the sum is composite. Their difference  $|a - b|$  is a positive multiple of 4, hence at least  $4 > 2$ , and even, hence composite. The thresholds in (??) are met with room to spare.  $\square$

**Proposition 4.** For every  $N \geq 1$ ,  $|A \cap [1, N]| = \lfloor N/4 \rfloor$ , i.e.  $N \leq 4|A \cap [1, N]| + 3$ .

*Proof.* The map  $i \mapsto 4(i + 1)$  is a bijection between  $\{0, \dots, k - 1\}$  and the  $k$  multiples of 4 in  $[1, 4k]$ , so  $|A \cap [1, 4k]| = k$ ; monotonicity extends this to  $N \leq 4|A \cap [1, N]| + 3$  with  $k = \lfloor N/4 \rfloor$ .  $\square$

### 4 Remark: $1/4$ appears to be the residue optimum

We searched systematically for periodic residue constructions denser than  $1/4$ . Exhaustively over all  $2^m$  subsets for every modulus  $m \leq 10$  (testing the realized sets  $A \cap [1, N]$  at  $N = 150$  over all distinct pairs), the maximum density attained is exactly  $1/4$ , e.g.  $m = 8$ ,  $S = \{0, 4\}$ . Greedy extension from the multiples of 4 together with 200 random trials per modulus for  $m \in \{12, 16, 20, 24, 28, 32, 40, 44, 48\}$  (tested at  $N = 200$ – $250$ ) never exceeded  $1/4$  either. We have not proved optimality, and record this only as computational evidence. It suggests the repaired statement, if the upper bound is true at all, should read  $|A| \leq (1/4 + \varepsilon)N$  with  $1/4$  attained by the multiples of 4 — not  $1/3$  by  $\{\pm 1 \bmod 6\}$ .

### 5 Formal verification

The accompanying Lean 4 project formalizes the refutation against Mathlib, using Mathlib’s `Nat.Prime` throughout.

- `claimed_extremal_invalid` — Theorem ??: the explicit witnesses  $5, 7 \in B$  with `Nat.Prime (7 - 5)`, discharged by `decide`.
- `inA_sum_admissible / inA_diff_admissible` — Proposition ??: sums of distinct multiples of 4 are even and  $> 2$ ; differences (for  $a < b$ ) are even and  $> 2$ ; both via an explicit divisibility witness closed by `omega`, composed with `not_prime_of_dvd`.
- `countA_lower` — Proposition ?? in the explicit form  $N \leq 4 \text{countA } N + 3$ , from the order-preserving enumeration `enumA` shown injective and landing in  $[1, 4k]$ .
- `conjecture_00000000036_false` — the conjunction: the claimed example fails, the multiples of 4 are admissible, and they have density  $1/4$ .

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000036_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.